

ECS7040P: Statistical Foundations for Artificial Intelligence

Introduction to Probability

Nivedita Viswanathan, PhD
Lecturer in Mathematics,
Queen Mary University of London
n.viswanathan@qmul.ac.uk

Overview

- ❑ Uncertainty in life : *How does probability theory help?*
- ❑ Probability: *What does it even mean?*
- ❑ Probability of events:
 - ❑ Union and Intersection of events
 - ❑ Mutually exclusive and exhaustive events
- ❑ Conditional Probability
 - ❑ Independent and Dependent events
 - The Case of Sally Clark
 - The Case of Lucia de Berk
 - ❑ Bayes' Theorem

Uncertainty in Life

How certain are we, about different things in life around us?

- **Will it rain tomorrow?** *“Weather prediction says it certainly will!”*
- **Will I be able to score well in the exams?** *“Likely, if I prepare well!”*
- **Will I get to class on time?** *“Very likely, if I leave early from home!”*
- **Will Djokovic win another grand slam in his career?** *In his own words, “Difficult, but not impossible!”*
- **“Will my sample dataset give me enough information about my population?”**

Some events are **more likely** to happen than others!

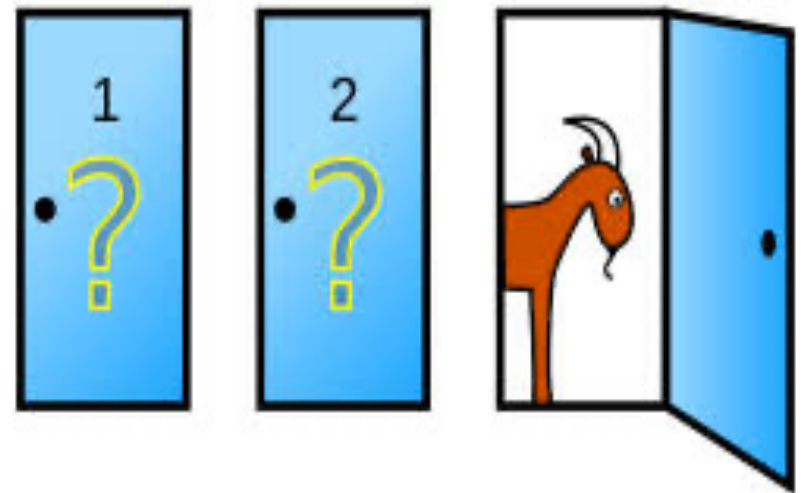
The Monty Hall Problem

- ❑ There are 3 doors, behind which there are 2 goats and 1 car.
- ❑ You pick a door. (You hope it has the car of course!)

Monty Hall is the game show host. Out of other 2 doors to open, the host chooses to open a door with goat behind.

- ❑ You can now choose to switch to the other unopened door.

What should you do?



The Birthday Paradox

In your class of approx., 50 people, how likely is it for a pair of you to share a common birthday?

I say, it is very likely!

❑ In fact, there is approximately 97% chance that 2 of you share a birthday.

Some very interesting probability problems: [Here](#)

What is probability?

- ❑ Probability is a way of **measuring uncertainty** about “something”.
- ❑ It is based on what we know/ believe/ assume.

How likely is it to rain tomorrow?

“The Weather App tells me there is an 80% chance of it raining!”

Will I be able to score an A in the exams?

“If I prepare well, I strongly believe that I have a 99.9% chance”.

Will Djokovic win another grand slam in his career?

“I would very much like to believe that he has 80% chance of another slam win!”

Vocabulary needed

- ❑ **Random experiment:** a process by which we observe something uncertain.
- ❑ **Outcome:** Result of a random experiment.
- ❑ **Sample Space:** Set of all possible outcomes.

Example: You are tossing a coin and the coin lands flat on the table.

Random experiment: Tossing the coin.

Outcomes: Head H or Tail T

Sample Space: $S = \{H, T\}$

❑ **Event:** Collection of possible outcomes. Say event A.

Note: It is **not all** possible outcomes. Just “a collection” of possible outcomes.

In set theory, this is called “subset of sample space”.

❑ **Complementary event:** Collection of possible outcomes **not in** A.

In the example before,

Experiment: Tossing a coin.

Events: A = Landing with a Head

= {H}

Complement of A = {T}

B = Landing with a Tail

= {T}

Complement of B = {H}

For Example: Suppose you roll a 6-faced die, and it stops with a face up.

Experiment: Rolling a 6-faced die.

Outcomes: 1, 2, 3, 4, 5 or 6

Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$

Events:

More examples

☐ Will it rain tomorrow?

Experiment:

Outcome:

Sample Space:

Event:

☐ The Birthday Paradox:

Experiment:

Outcome:

Event:

Sample Space:

Probability of an event

Given a **random experiment**, we want

Probability of an event

Suppose A is an event, we say

Probability of an event A is

For example: Suppose you roll a die.

Let event A: die stopping with 2 on its face.

Probability of event A = Probability of the die stopping with 2 on its face

We denote it as **$P(A)$** .

Read it as “Probability of event A”.

How do you measure uncertainty?

Probability **measures uncertainty**.

Always measured **on a scale of 0 to 1**.

- ❑ If the event A is **impossible** to occur, $P(A) = 0$.
- ❑ If the event A is **certain to occur**, $P(A) = 1$.
- ❑ If the event is **neither known to be impossible or certain to occur**,

$$0 < P(A) < 1$$

Event is impossible.
For ex: Rolling a 7 on
a 6-faced die that
has numbers 1 - 6

0

0.5

1

Event is certain to
happen.
For ex: Tomorrow
comes after
today!

Equal chance of event
happening or not.

How do you compute probability?

Probability of an event $A = P(A) = \frac{\text{Number of ways in which event } A \text{ can happen}}{\text{Number of elements in the sample space}}$

For example: Suppose you roll a 6-faced die.

What is the probability of the die stopping with 2 on top of its face?

Example

A: event that die stops with 2 on its face.

Probability of die stopping with 2 on its face = $P(A)$

$$P(A) = \frac{\text{Number of ways it can stop with 2 on its face}}{\text{Number of elements in Sample Space}}$$

$$= \frac{1}{6}$$

Complementary events

Suppose A: event

we denote the complementary event of A : A'

Example: Suppose you roll a six-faced die.

A: the event that the die stops with 2 on its face.

A': the event that the die **does not stop** with 2 on its face.

But this means, A' : the event that the die stops with 1/3/4/5/6 on its face.
(We do know that the die stops)

$$P(A') = \frac{5}{6} = 1 - \frac{1}{6} = 1 - P(A) \text{ (we will come to this)}$$

Odds of an event

The odds of an event A is defined by

$$\frac{P(A)}{P(A')} = \frac{P(A)}{1-P(A)}$$

That is, odds of an event tells us

how much more likely it is that the event happens than that it does not happen!

For example: If $P(A) = \frac{2}{3}$, $1-P(A) = \frac{1}{3}$.

$$\text{Odds of A} = \frac{2}{1}.$$

Odds of event A happening is “2 to 1” in favour of A.

The Case of Lucia de Berk

- 2003: Lucia de Berk was sentenced to life imprisonment (Netherlands), for 7 counts of murders and 3 counts on attempted murders.
- Deaths during her shifts
- Statistician: the odds were 342 million-to-one against that it was a coincidence
- 2008 Released, 2010 acquitted



Court clears Dutch nurse jailed for murdering patients

Lucy de Berk, who served six years of life sentence for murders of seven patients, exonerated after retrial hears of experts' errors



Lucy de Berk, right, after being cleared at a retrial in Arnhem, Netherlands, of the murder of seven patients in her care. Photograph: AP

https://en.wikipedia.org/wiki/Lucia_de_Berk_case

<https://www.theguardian.com/world/2010/apr/14/dutch-nurse-murder-patients-cleared>

How to understand the odds here?

Let

event A : the deaths occurred by chance when Lucia was present

event A' : the deaths **did not occur** by chance when Lucia was present (that is, she is a murderer)

$$\frac{P(A)}{P(A')} = \frac{1}{342,000,000}$$

Axioms of Probability

Given a random experiment E, sample space S and an event A

Axiom 1:

$$0 \leq P(A) \leq 1$$

— Always between

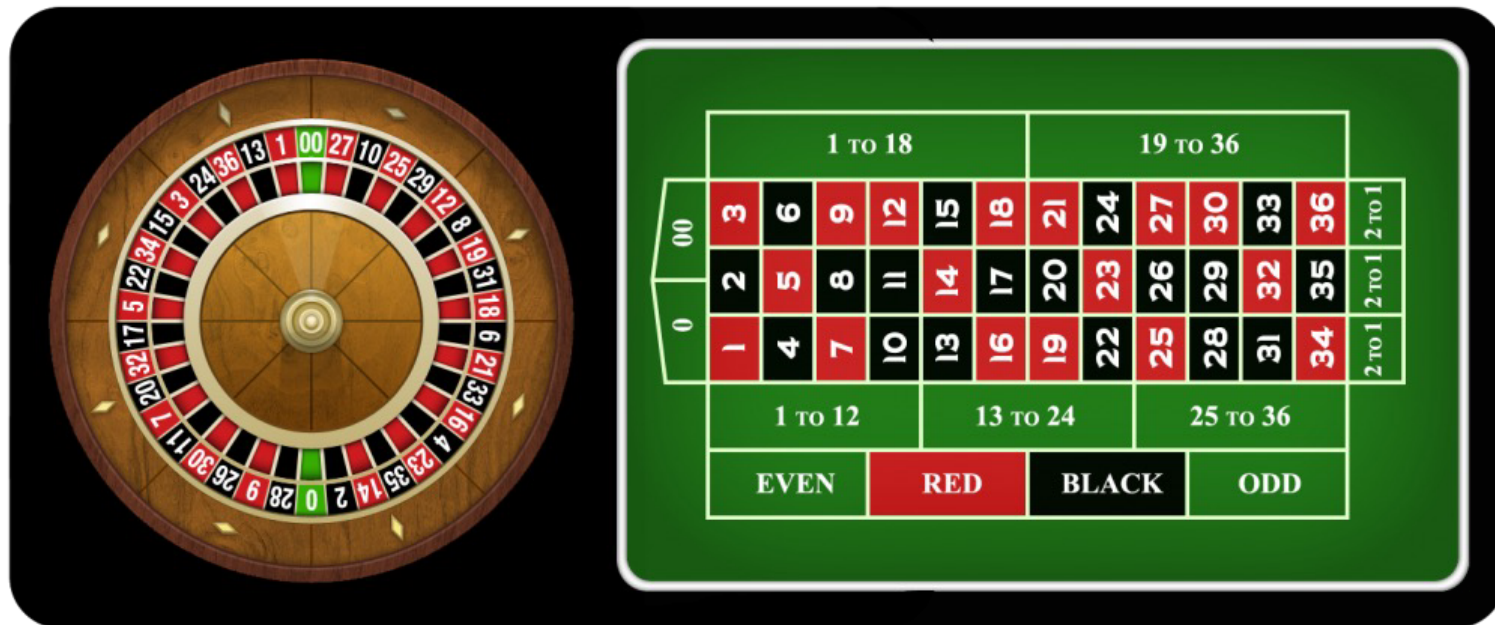
0 — 1

Axiom 2:

$$P(S) = 1$$

Example: Roulette Board

You spin the wheel! Ball falls into a pocket. Let's play!



This Photo by Unknown Author is licensed under [CC BY-SA-NC](#)

Probability of events

What is the probability of the following events?

$P(\text{ball falling in a red pocket}) = ?$

0	00		3	6	9	12	15	18	21	24	27	30	33	36
			2	5	8	11	14	17	20	23	26	29	32	35
			1	4	7	10	13	16	19	22	25	28	31	34

$P(\text{ball falling in an even pocket}) = ?$

Union of events

$P(\text{ball falling in a red or even pocket}) = ?$

A: ball falling in a red pocket

B: ball falling in an even pocket

C: event A or B

$P(C) =$

0	00	3	6	9	12	15	18	21	24	27	30	33	36
	2	5	8	11	14	17	20	23	26	29	32	35	
	1	4	7	10	13	16	19	22	25	28	31	34	

Represented as $P(A \cup B)$

Read as **Probability of event A or event B.**

Is it equal to $P(\text{ball falling in a red pocket}) + P(\text{ball falling in an even pocket}) = P(A) + P(B)$?

$P(\text{ball falling in a red or black pocket}) = ?$

A: ball falling in a red pocket

B: ball falling in a black pocket

C: event A or B

$P(C) =$

0	00	3	6	9	12	15	18	21	24	27	30	33	36
	2	5	8	11	14	17	20	23	26	29	32	35	
	1	4	7	10	13	16	19	22	25	28	31	34	

Is it equal to $P(\text{ball falling in a red pocket}) + P(\text{ball falling in a black pocket}) = P(A) + P(B)$?

Intersection of events

$P(\text{ball falling in a red and even pocket}) = ?$

A: ball falling in a red pocket

B: ball falling in an even pocket

C: event A and B

$P(C) =$

0	00		3	6	9	12	15	18	21	24	27	30	33	36
	2	5	8	11	14	17	20	23	26	29	32	35		
	1	4	7	10	13	16	19	22	25	28	31	34		

Represented as $P(A \cap B)$

Read as **Probability of event A and event B.**

Is it equal to $P(\text{ball falling in a red pocket}) + P(\text{ball falling in an even pocket})$?

Intersection of events

$P(\text{ball falling in a red and black pocket}) = ?$

A: ball falling in a red pocket

B: ball falling in a black pocket

C: event A and B

$P(C) =$

0	00		3	6	9	12	15	18	21	24	27	30	33	36
	2	5	8	11	14	17	20	23	26	29	32	35		
	1	4	7	10	13	16	19	22	25	28	31	34		

Is it equal to $P(\text{ball falling in a red pocket}) + P(\text{ball falling in a black pocket})$?

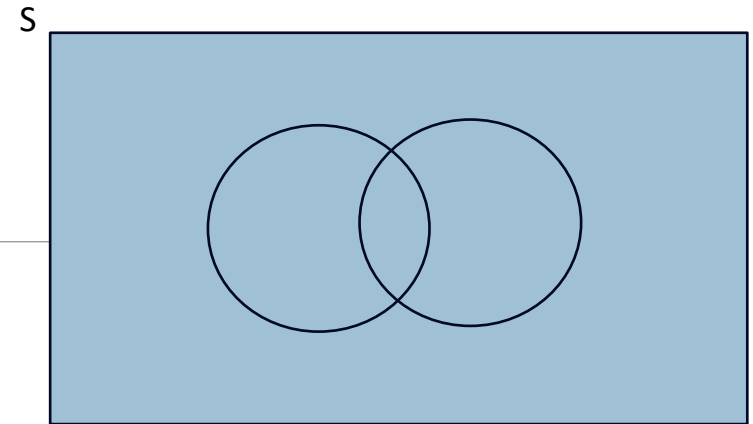
Visualising sets

- ❑ [Venn Diagrams](#) (named after John Venn)
- ❑ Used to represent sets (collection of things)
- ❑ How to draw it? (for probability theory)

- ❑ The outer rectangular box: Sample Space S
- ❑ For every event, draw a circle.
- ❑ Do the circles meet?

Depends on whether the events intersect.

- ❑ What to write in it?
 - ❑ Inside circle: Number of ways event can happen
 - ❑ Inside rectangle: Total number of elements in sample space



Note: Size does not matter!

Venn Diagrams: Example

What is the Venn Diagram for

A : ball falling in red pocket

B: ball falling in even pocket

From it, can you find $P(A)$; $P(B)$; $P(A \cap B)$; $P(A \cup B)$?

0		00
1	2	3
4	5	6
7	8	9
10	11	12
13	14	15
16	17	18
19	20	21
22	23	24
25	26	27
28	29	30
31	32	33
34	35	36

Venn Diagrams: Example

What is the Venn Diagram for

A : ball falling in red pocket

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0		00
1	2	3
4	5	6
7	8	9
10	11	12
13	14	15
16	17	18
19	20	21
22	23	24
25	26	27
28	29	30
31	32	33
34	35	36

Union Rule

For 2 events A and B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Do you see why?

Mutually exclusive events

Events A and B in sample space S are said to be **mutually exclusive** if

$$P(A \cap B) = 0.$$

When events are mutually exclusive,

$$P(A \cup B) = P(A) + P(B)$$

For example, $S = A \cup A'$ and $P(A \cap A') = 0$.

Above rule for mutually exclusive events, along with Axiom 2, means $P(A') = 1 - P(A)$

Exhaustive events

Given a sample space S , the set of events that taken together make up the entire sample space are called **exhaustive events**.

That is, suppose S is the sample space and A , B and C are events such that

$$A \cup B \cup C = S.$$

Then A , B , C **together** are exhaustive events.

Example: Suppose you roll a 6-faced die.

Sample Space $S = \{1, 2, 3, 4, 5, 6\}$ Events: $A = \{1, 3, 5\}$; $B = \{2, 4, 6\}$

$$S = A \cup B$$

Then A and B together are exhaustive events.

Roulette Example

□ Given the roulette game, which set of events below are exhaustive?

A: ball falling in a black pocket

B: ball falling in an even pocket

C: ball falling in a green pocket

D: ball falling in an odd pocket

0		00
1	2	3
4	5	6
7	8	9
10	11	12
13	14	15
16	17	18
19	20	21
22	23	24
25	26	27
28	29	30
31	32	33
34	35	36

Conditional Probability

Suppose I tell you that the ball falls in a red pocket.

Now, how likely is it to have been the 19 of red?

Draw Venn Diagrams and check if that helps.

0		00
1	2	3
4	5	6
7	8	9
10	11	12
13	14	15
16	17	18
19	20	21
22	23	24
25	26	27
28	29	30
31	32	33
34	35	36

Conditional Probability

Let events

B: ball falls in a red pocket.

A: ball falls into pocket numbered 19.

$P(\text{A given B})$ = probability of event A **given** event B happens

$$= P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Here, of course, you need $P(B) \neq 0$.

0		00
1	2	3
4	5	6
7	8	9
10	11	12
13	14	15
16	17	18
19	20	21
22	23	24
25	26	27
28	29	30
31	32	33
34	35	36

Example

Find $P(\text{ball falls in black pocket} \mid \text{ball falls in odd numbered pocket})$.

Find $P(\text{ball falls in red pocket} \mid \text{ball falls in green pocket})$

0		00
1	2	3
4	5	6
7	8	9
10	11	12
13	14	15
16	17	18
19	20	21
22	23	24
25	26	27
28	29	30
31	32	33
34	35	36

Independent Events

Two events A and B are **independent** if
the probability of event B happening is the same, **whether or not** event A has occurred.

For example, suppose you play the roulette twice.

B: ball falls in a red pocket the first time

A: ball falls in a black pocket the second time.

What is $P(A|B) = ?$

Two events A and B are said to be independent when

$$P(A|B) = P(A)$$

Event B happening **does not affect** the probability of event A happening.

Recall

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

From this we get the **multiplication rule** when events A and B are independent,

$$P(A \cap B) = P(A)P(B)$$

Adding and Multiplying probabilities

Given events A and B in a random experiment.

If A and B are **mutually exclusive**, then

$$P(A \cup B) = P(A) + P(B)$$

If A and B are **independent events**, then

$$P(A \cap B) = P(A) P(B)$$

The Case of Sally Clark

- 1999: Mrs Clark convicted of double murder (of her babies)
- Sudden Infant Dead Syndrome (SIDS)
- Eminent paediatrician Professor Sir Roy Meadow:
"1 in 73 million" (8543×8543)

Double murder Vs double SIDS



Mrs Clark, 38, from Wilmslow, Cheshire, was convicted of murder at Chester Crown Court in November 1999, wrongly convicted, found dead by her family at her home (2003).

<http://news.bbc.co.uk/1/hi/england/2698425.stm>

The Case of Sally Clark

What was considered?

Data: $P(\text{an infant dying of SIDS}) = \frac{1}{8543}$

Assumption: 2 infants in the same household dying of SIDS are independent events.

$$\begin{aligned} &P(2^{\text{nd}} \text{ infant dying of SIDS and } 1^{\text{st}} \text{ infant died of SIDS}) \\ &= P(2^{\text{nd}} \text{ infant dying of SIDS}) \times P(1^{\text{st}} \text{ infant dying of SIDS}) \end{aligned}$$

What is the mistake in this argument?

$P(2 \text{ infants dying of SIDS}) \neq P(1^{\text{st}} \text{ infant dying of SID}) \times P(2^{\text{nd}} \text{ infant dying of SID})$

Why?

The 2 children belong to the same household. They were born to the same parents.

So why should their health symptoms be independent of each other?

2nd infant dying of SIDS is **not independent** of 1st infant dying of SIDS

The Case of Lucia de Berk

- 2003: Lucia de Berk was sentenced to life imprisonment (Netherlands), for 7 counts of murders and 3 counts on attempted murders.
- Deaths during her shifts
- Statistician: the odds were 342 million-to-one against that it was a coincidence
- 2008 Released, 2010 acquitted

What was the error?

Deaths at different hospitals taken as independent events. Read here : [The Science](#)



Court clears Dutch nurse jailed for murdering patients

Lucy de Berk, who served six years of life sentence for murders of seven patients, exonerated after retrial hears of experts' errors



Lucy de Berk, right, after being cleared at a retrial in Arnhem, Netherlands, of the murder of seven patients in her care. Photograph: AP

https://en.wikipedia.org/wiki/Lucia_de_Berk_case

<https://www.theguardian.com/world/2010/apr/14/dutch-nurse-murder-patients-cleared>

Law of Total Probability

Given E_1 and E_2 mutually exclusive and exhaustive events.

That means,

$$P(E_1 \cap E_2) = 0 \text{ and } E_1 \cup E_2 = S.$$

Any other event A can happen either along with E_1 or along with E_2 .

$$A = (A \cap E_1) \cup (A \cap E_2)$$

Probability of any other event A happening

$$P(A) = P(A \cap E_1) + P(A \cap E_2)$$

Example

Suppose you roll a 6-faced die that stops with a face up. $S = \{1, 2, 3, 4, 5, 6\}$

Let event A = die stops with an even number event B = die stops with an odd number

$$A = \{2, 4, 6\}$$

$$B = \{1, 3, 5\}$$

Any event now (die stopping with any number on its face up), is either an even or an odd number.

For ex: Consider the event of die stopping with a number 2 on its face up.

That is, $E = \{2\}$

Since $A \cup B = S$,
$$E = (E \cap A) \cup (E \cap B)$$



Bayes' Theorem

□ Who is this gentleman?

You can read about this contested picture: [HERE](#).

About Bayes' Theorem: [HERE](#).

□ Wikipedia says

“Bayes' Theorem gives a mathematical rule for inverting conditional probabilities, allowing the probability of a cause to be found given its effect.”

Statement of Bayes' Theorem

Given A and B: events in the sample space of an experiment

$$\begin{aligned} P(B|A) &= \frac{P(B \cap A)}{P(A)} \\ &= \frac{P(A|B)P(B)}{P(A)} \\ &= \frac{P(A|B)P(B)}{P(A \cap B) \cup P(A \cap B')} \\ &= \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')} \end{aligned}$$

Example

A laboratory blood test is as follows.

- ❑ 95% effective in detecting a certain disease when it is in fact, present.
- ❑ Yields a “false positive” result for 1% of the healthy persons tested.

If 0.5% of the population has the disease, what is the probability that
a person has the disease
given
that the test result is positive?

Experiment:

Sample Space:

Events:

A: person has the disease

B: the test result showing positive

$$P(A) = 0.5/100$$

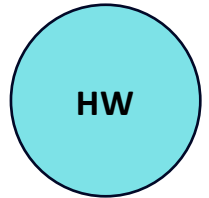
$$P(A') = 99.5/100$$

$$P(B|A') = 1/100$$

$$P(B|A) = 95/100$$

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(B|A)P(A)}{P(B)} \\ &= \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A')P(A')} \\ &= \frac{(0.95)(0.005)}{(0.95)(0.005) + (0.01)(0.995)} \\ &= 0.323 \end{aligned}$$

Thus, only 32% of the time when the test results are positive the person actually has the disease.



All probabilities are conditional!

- ❑ Two very different approaches to measuring uncertainty :

frequentist(what we have seen) and subjective (Bayesian statistics)

To understand the statement “All probabilities are conditional” better ,

- ❑ Read Chapter 6,

“Risk Assessment and Decision Analysis with Bayesian Networks”

by Norman Fenton.



Queen Mary
University of London